

Trainings ▾

Preparatory Steps

with Electronically Actuated Injector

- Remove the crankcase breather. Refer to Procedure 003-001 in Section 3
(/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html).
- Remove the right and left bank engine harness. Refer to Procedure 019-043 in Section 19
(/qs3/pubsys2/xml/en/procedures/56/56-019-043-tr.html).
- Remove the ECM cooling plate. Refer to Procedure 008-128 in Section 8
(/qs3/pubsys2/xml/en/procedures/56/56-008-128.html).
- Remove the fuel manifold (drain). Refer to Procedure 006-021 in Section 6
(/qs3/pubsys2/xml/en/procedures/56/56-006-021-tr.html).
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5 (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html).
- Remove the engine speed sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-042 in Section 19
(/qs3/pubsys2/xml/en/procedures/19/19-019-042.html)
- Remove the intake manifold temperature and pressure sensors. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-059
(/qs3/pubsys2/xml/en/procedures/19/19-019-059.html) and Refer to Procedure 019-061 in Section 19
(/qs3/pubsys2/xml/en/procedures/19/19-019-061.html).
- Remove the low temperature coolant supply and bypass tubes and the water jacket coolant bypass tube. Refer to Procedure 008-062 in Section 8



(/qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html).

- If equipped with a Centinel™ system, remove the line from the sentinel valve to the fuel drain. Use the following procedure in the Centinel™ Master Repair Manual, Bulletin 3666231. Refer to Procedure 007-088 in Section 7. (/qs3/pubsys2/xml/en/procedures/96/96-007-088-tr.html)

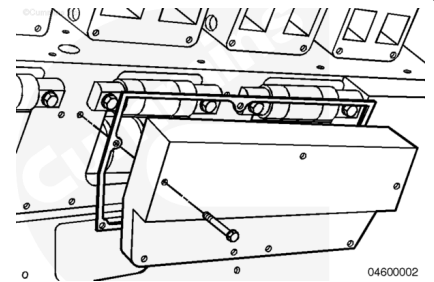
Remove

with Mechanically Actuated Injector

Remove the cam follower cover capscrews and the cam follower cover.

Remove and inspect the cam follower gasket.

Discard the gasket **only** if damaged.

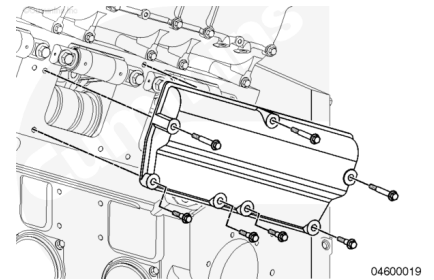


with Electronically Actuated Injector

Remove the capscrews securing the cam follower covers to the engine.

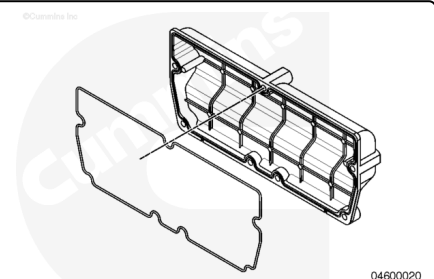
Remove the cam follower cover.

Note the location of the different lengths of the capscrews.



Remove the gasket from the cam follower cover.

Save the gasket for possible reuse.



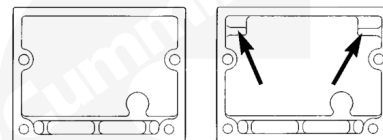
Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the cam follower cover using solvent that will **not** damage aluminum.

Dry with compressed air.

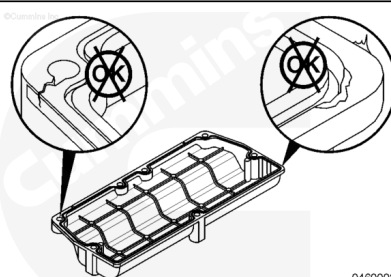
Check the cam follower cover for cracks or other damage to the sealing surface.

with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

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⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam to clean the cam follower covers.

Dry with compressed air.

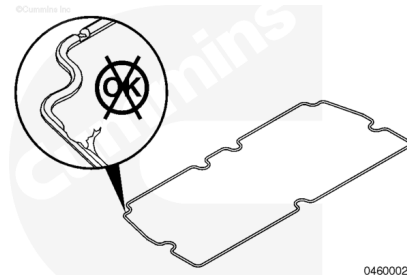
Do **not** use steam or cleaning solvents on the gasket. Use a clean cloth and wipe the gasket clean.

Inspect the cam follower covers for nicks, burrs, cracks, or other damage.

Use a fine crocus cloth to remove small nicks and burrs.

Inspect the threaded capscrew holes for damage. Damaged holes can be repaired using Cummins® screw thread insert, Part Number 3822700 (M10x1.5-6) or equivalent.

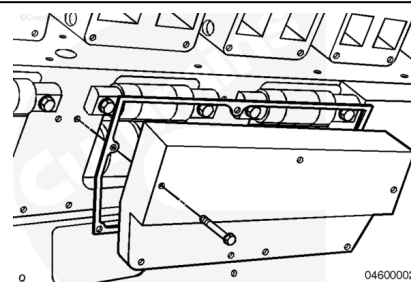
Inspect the cam follower gasket for signs of cracks, tears, or other damage. If damage is found, replace the gasket.



Install

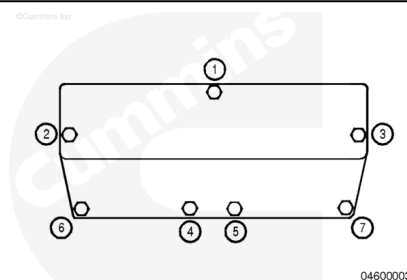
with Mechanically Actuated Injector

Install the gasket, cam follower cover, and capscrews.



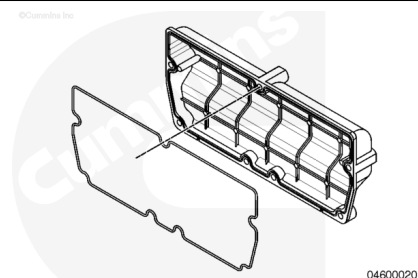
Tighten the seven capscrews in the sequence shown in the illustration.

Torque Value: 45 n•m [33 ft-lb]



with Electronically Actuated Injector

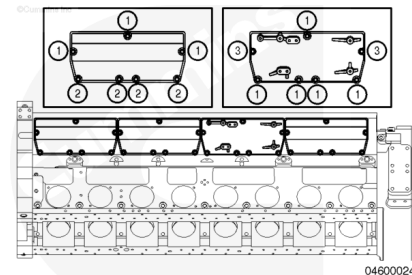
Install the gasket into the cam follower cover.



Install the left bank cam follower covers in the position shown.

Note the locations for the different length capscrews.

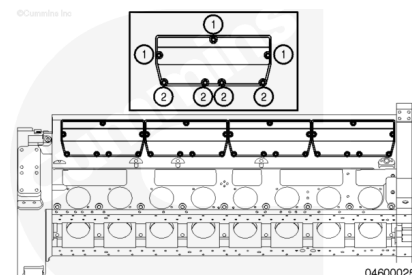
1. M10 x 80
2. M10 x 45
3. M10 x 65.



Install the right bank cam follower covers in the position shown.

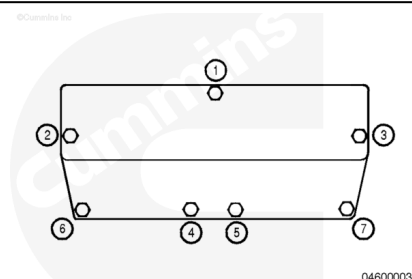
Note the locations of the different length capscrews.

1. M10 x 80
2. M10 x 45



Tighten the capscrews in the sequence shown in the illustration.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

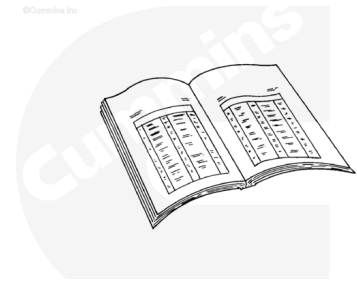
with Electronically Actuated Injector

- Install the low temperature coolant supply and bypass tubes and the water jacket coolant bypass tube. Refer to

Procedure 008-062 in Section 8

(/qs3/pubsys2/xml/en/procedures/56/56-008-062.html).

- Install the fuel manifold (drain). Refer to Procedure 006-021 in Section 6
(/qs3/pubsys2/xml/en/procedures/56/56-006-021-tr.html).
- Install the fuel pump. Refer to Procedure 005-016 in Section 5 (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html).
- Install the engine speed sensor. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-042 in Section 19
(/qs3/pubsys2/xml/en/procedures/19/19-019-042.html)
- Install the intake manifold temperature and pressure sensors. Use the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-059
(/qs3/pubsys2/xml/en/procedures/19/19-019-059.html) and Refer to Procedure 019-061 in Section 19
(/qs3/pubsys2/xml/en/procedures/19/19-019-061.html)
- Install the ECM cooling plate. Refer to Procedure 008-128 in Section 8 (/qs3/pubsys2/xml/en/procedures/56/56-008-128.html).
- Install the right and left bank engine harness. Refer to Procedure 019-043 in Section 19
(/qs3/pubsys2/xml/en/procedures/56/56-019-043-tr.html).
- Install the crankcase breather. Refer to Procedure 003-001 in Section 3 (/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html).
- If equipped with the Centinel™ System, install the line from the centinel valve to the fuel drain. Use the following procedure in the Master Repair Manual Centinel™ System, Bulletin 3666231. Refer to Procedure 007-088 in Section 7
(/qs3/pubsys2/xml/en/procedures/96/96-007-088-tr.html)



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